

Ioannis-Aris Koufopoulos

MASTER'S GRADUATE · DATA SCIENCE

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Education

University of Piraeus

Piraeus

MSC IN CYBERSECURITY AND DATA SCIENCE

Oct 2022 - Jul 2024

- Specialization: Data Science
- Thesis: Fact-Checking with LLMs [PDF]
- Advisor: Dr. George Papastefanatos (Athena RC)

Athens University of Economics and Business

Athens

BSc IN INFORMATICS

Oct 2017 - Sep 2022

- Specializations: Computer Systems and Networks - Information systems and security

Professional experience

Research associate

Oct 2024-Present

ATHENA RESEARCH CENTER

- Built and optimized Retrieval-Augmented Generation (RAG) pipelines, including data ingestion, embeddings, vector search, and retrieval optimization.
- Constructed knowledge graphs from unstructured text using entity extraction and relationship modeling.
- Applied large language models (LLMs) for question answering, summarization, and text generation tasks.
- Developed machine learning and NLP models for text classification and semantic understanding.
- Collected, cleaned, and processed large-scale web text data to support RAG systems and model training.

Publications

Georgios Kaoukis, Ioannis-Aris Koufopoulos, Eleni Psaroudaki, Danae Pla Karidi, Evaggelia Pitoura, George Papastefanatos, and Panayiotis Tsaparas. Emeraldmind: A knowledge graph-augmented framework for greenwashing detection. In Proceedings of the ACM Web Conference 2026 (WWW '26), page 11, New York, NY, USA, 2026. ACM. 7 [PDF] [Code]

Georgios Kaoukis, Ioannis-Aris Koufopoulos, Eleni Psaroudaki, Danae Pla Karidi, Evaggelia Pitoura, George Papastefanatos and Panayiotis Tsaparas. EmeraldApp: An AI-Driven Tool for Detecting Greenwashing in Sustainability Claims. The ACM Web Conference 2026 (WWW '26), The 6th International Workshop on Computational Methods for Online Discourse Analysis (BeyondFacts '26).

Skills

Programming Languages: Python, R, SQL, Java.

Tools and Frameworks: PyTorch, Hugging Face Transformers, spaCy, NLTK, scikit-learn, LangChain, Pandas, NumPy.

Specializations: Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Knowledge Graph Construction, LLM Evaluation and Prompt-Based Reasoning, Data Harvesting and Cleaning.

Key Projects

Fact Checking Tool: Built an end-to-end Retrieval-Augmented Generation (RAG) tool for tracking and evaluating false claims in multiple languages (Greek and English).

Text summarization tool: Developed a text summarization tool that converts large corpora into concise bulleted summaries using large language models (LLMs).

Synthetic dataset generation: Generated a synthetic dataset for greenwashing detection, with the utilization of LLMs and the extraction of news articles.

Awards, Fellowships, & Grants

2024 **MSc Academic Achievement Award**, University of Piraeus

2024 **Merit-Based Scholarship**, University of Piraeus

Languages

English: C2 Proficiency – ECPE Certificate (University of Michigan).

Greek: Native